CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

IT342-G2
SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: Mini App – User Registration & Authentication

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1. Introduction

1.1. Purpose

- This system is a mini application that allows users to register, log in, and log out. It ensures that only authenticated users can access protected pages, and that user data is stored securely. The intended audience for this document includes developers, testers, and stakeholders who need a clear understanding of the system functionality.

1.2. Scope

The system provides the following functionality:

- User registration
- User login and authentication
- User logout
- Access control for protected pages

The system does not include additional features like password recovery, email verification, or advanced dashboards. Its main focus is on providing a simple and secure authentication process.

1.3. Definitions, Acronyms, and Abbreviations

- JWT – JSON Web Token, used for user authentication
- UI – User Interface (React frontend)
- API – Application Programming Interface (Spring Boot backend)
- ERD – Entity Relationship Diagram
- DTO – Data Transfer Object

2. Overall Description

2.1. System Perspective

The mini app consists of three main components:

- Frontend (Web - React UI): Provides forms and buttons for user actions on web browsers.
- Frontend (Mobile - Android/Kotlin): Provides native mobile interface for user actions on Android devices.
- Backend (Spring Boot API): Handles business logic, authentication, and database interaction for both web and mobile clients.
- Database (MySQL): Stores user data securely.

Both web and mobile applications connect to the same backend API, ensuring consistent data and functionality across platforms.

2.2. User Classes and Characteristics

- Guest User: Can register and log in; cannot access protected pages.
- Authenticated User: Can log in and log out; has access to protected content (Dashboard and Profile pages).

2.3. Operating Environment

- Web Frontend: React JS, running in a browser
- Mobile Frontend: Android Studio, Kotlin language, running on Android devices (API level 24+)
- Backend: Spring Boot
- Database: MySQL
- Tools: Draw.io (diagrams), Android Studio (mobile development)

2.4. Assumptions and Dependencies

- Users must provide a valid email to register.
- The system relies on a stable connection between frontend (web/mobile), backend, and database.
- JWT is used for authentication on both web and mobile platforms.
- Passwords are securely hashed before storing in the database.
- Mobile application requires internet connection to communicate with backend API.

3. System Features and Functional Requirements

Describe each major feature of the system and its functional requirements.

3.1. Feature 1:User Registration

Description: Allows a new user to create an account using first name, last name, email, and password.
Functional Requirements:

- The system checks if the email is already registered.
- Passwords are hashed before saving to the database.
- Successful registration returns a confirmation message.
- Available on both web and mobile platforms.

3.2. Feature 2: User Login

Description: Allows registered users to log in using email and password.
Functional Requirements:

- The system validates the email and password.
- Successful login returns a JWT token for session management.
- Invalid credentials return an error message.
- JWT token is stored securely (localStorage for web, SharedPreferences for Android).
- Available on both web and mobile platforms.

3.3. Feature 2: User Logout

Description: Allows authenticated users to log out securely.
Functional Requirements:

- The system invalidates the JWT token.
- After logout, protected pages cannot be accessed.
- The user is redirected to the login screen.
- Available on both web and mobile platforms.

4. Non-Functional Requirements

- **Performance**: API responses should be under 5 seconds.
- **Security:** Passwords are hashed; JWT tokens expire after 24 hours.
- **Usability:** Simple and intuitive interface for registration, login, and logout.
- **Reliability:** Handles invalid inputs gracefully.
- **Maintainability:** Modular backend design allows for future enhancements.

Compatibility:

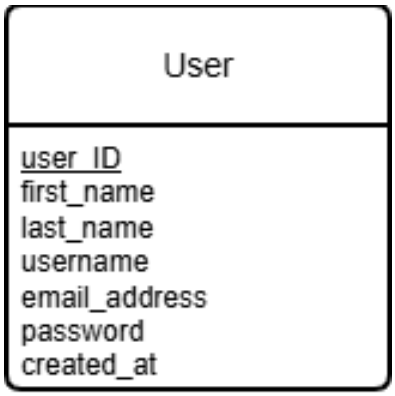
- Web: Compatible with modern browsers (Chrome, Firefox, Safari, Edge)
- Mobile: Compatible with Android devices.

Responsiveness:

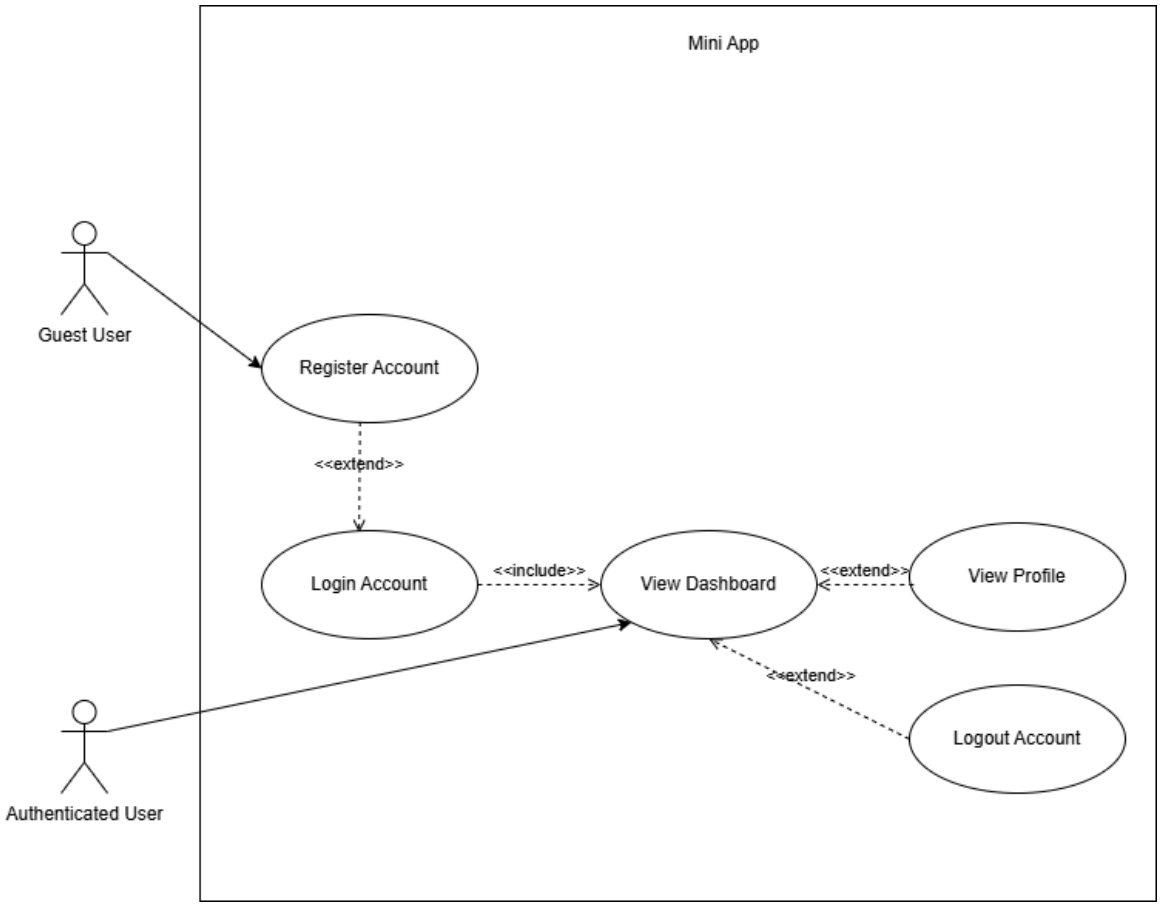
- Web: Responsive design adapts to different screen sizes
- Mobile: Android UI optimized for mobile devices

5. System Models (Diagrams)

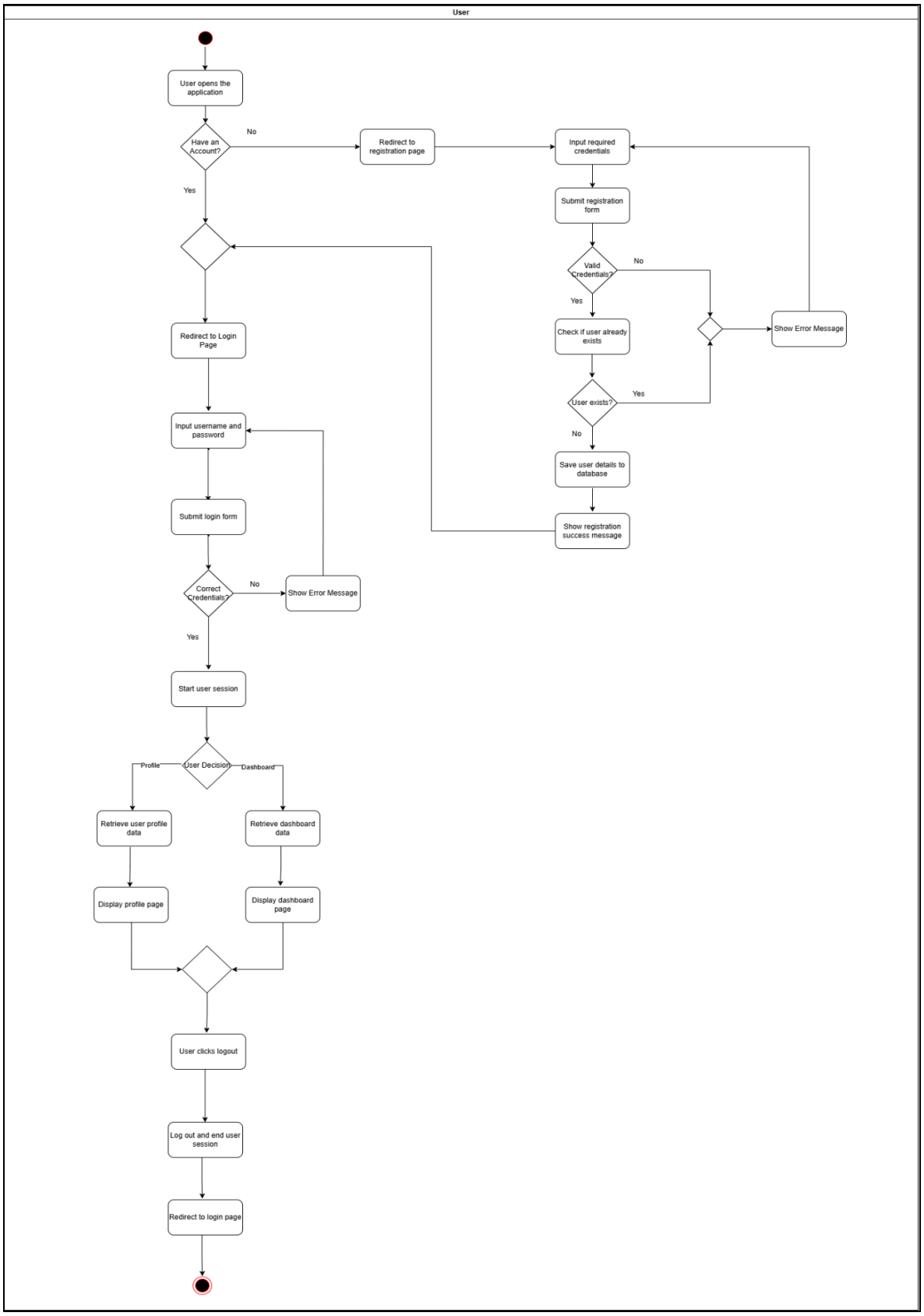
5.1. ERD



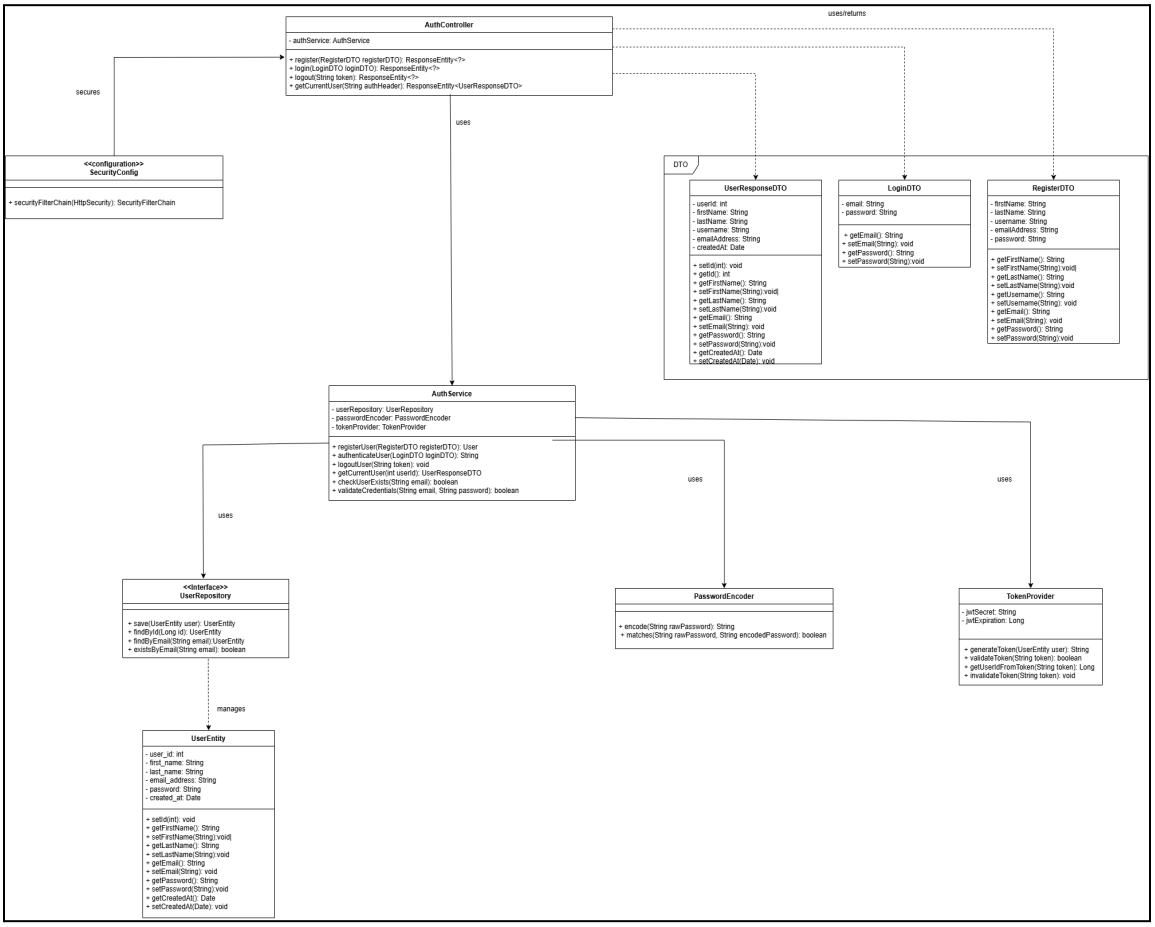
5.2. Use Case Diagram



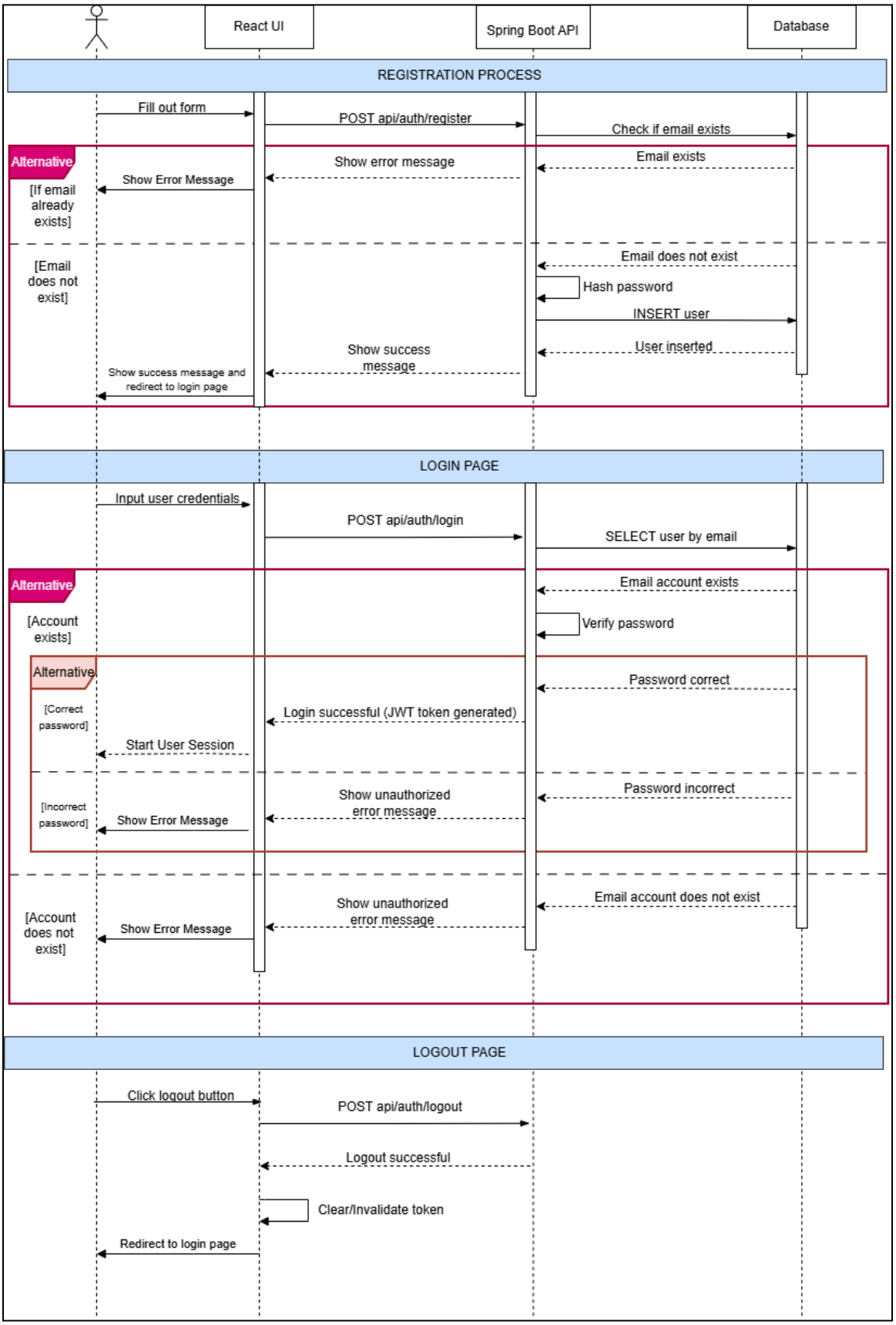
5.3. Activity Diagram



5.4. Class Diagram



5.5. Sequence Diagram

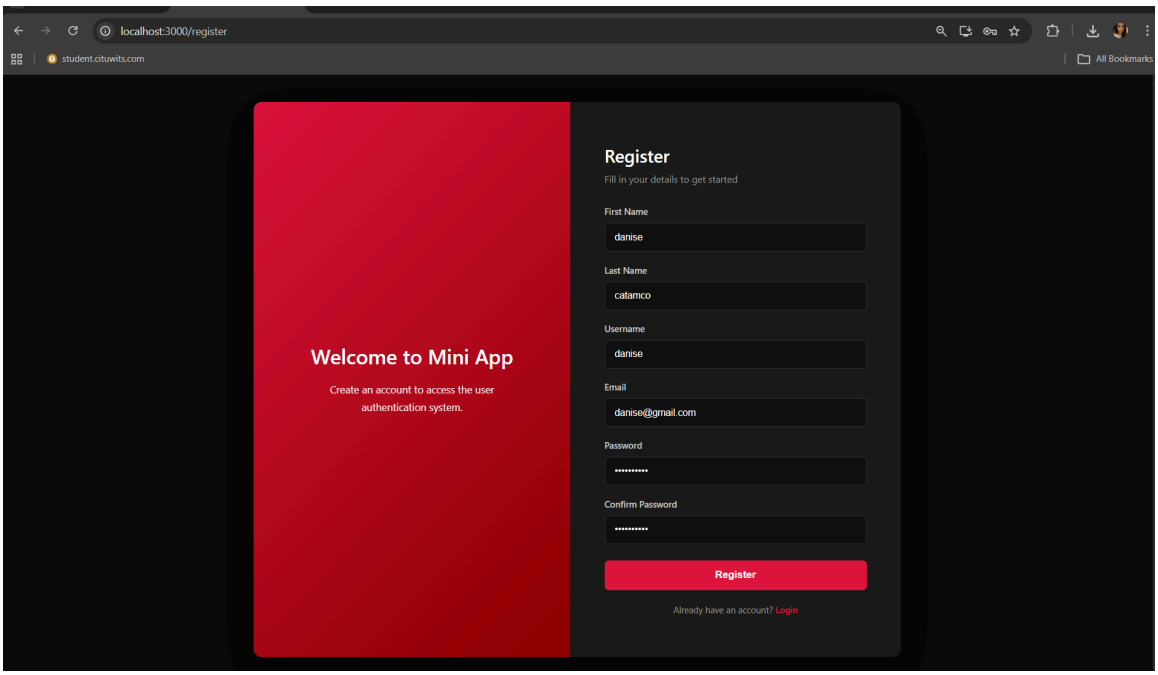


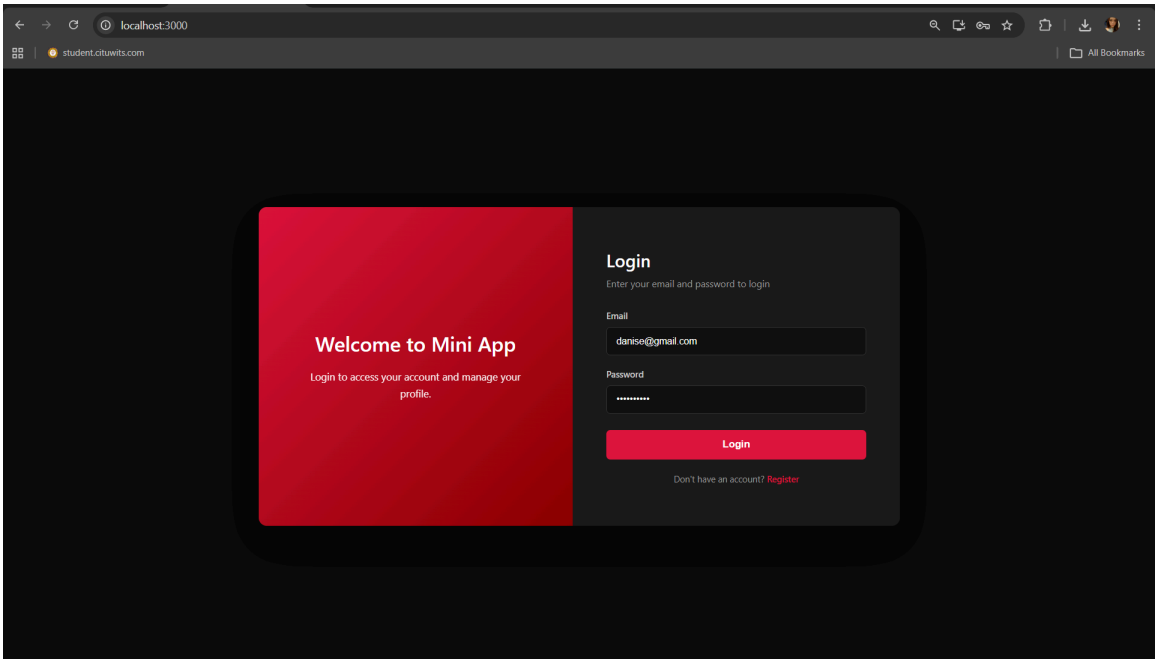
6. Appendices

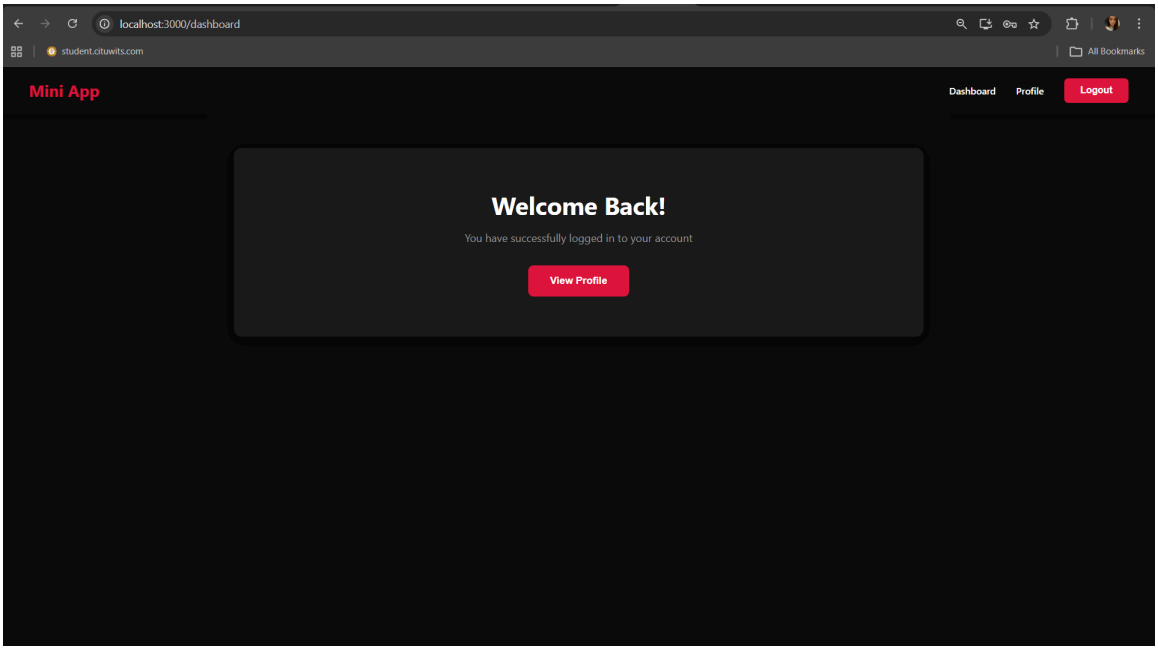
Tools used:

- Draw.io - Diagram creation tool
- Spring Boot - Backend framework (shared by web and mobile)
- React - Web frontend framework
- Android Studio - Mobile development IDE
- Kotlin - Mobile application programming language
- MySQL - Database management system
- Retrofit - Android HTTP client for API communication
- BCrypt - Password hashing library

Web Screenshots:



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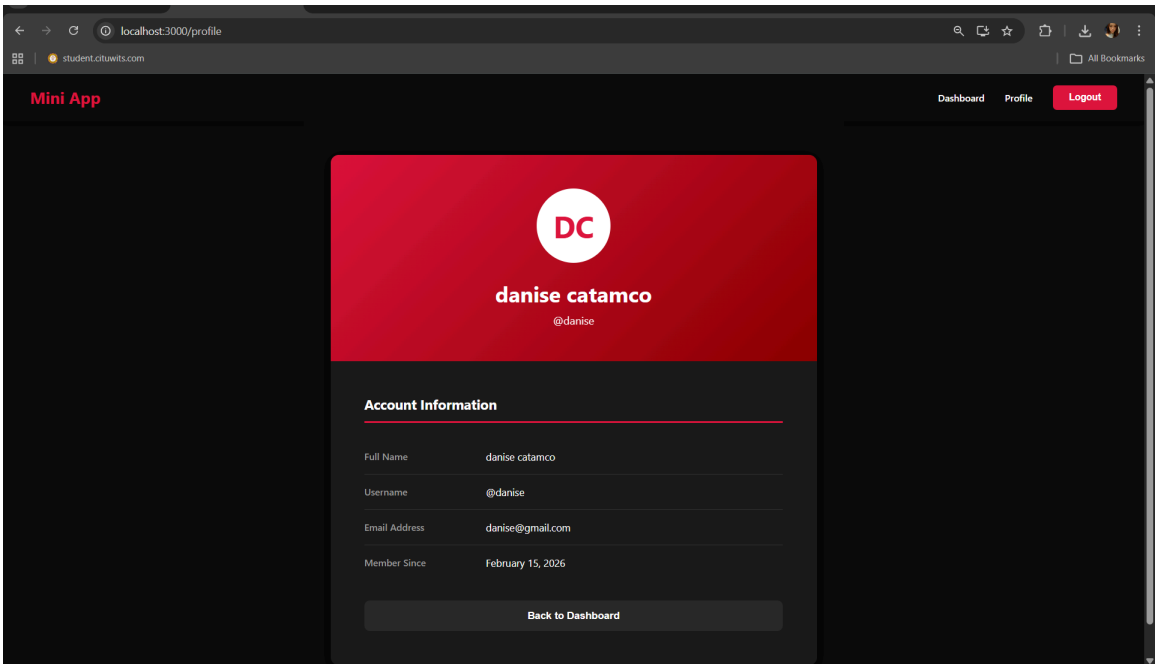
All Bookmarks
Mini App
Dashboard
Profile
Logout
Welcome Back!
You have successfully logged in to your account
View Profile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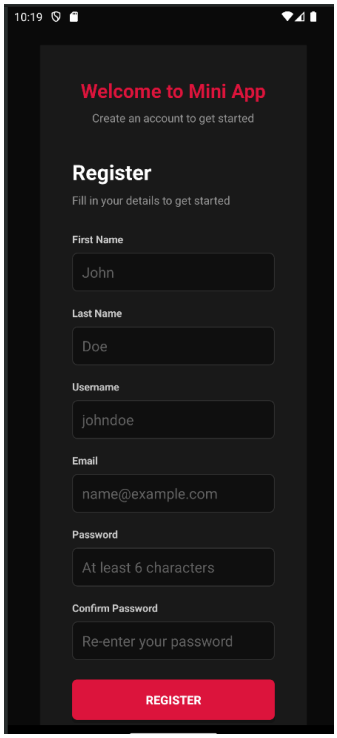
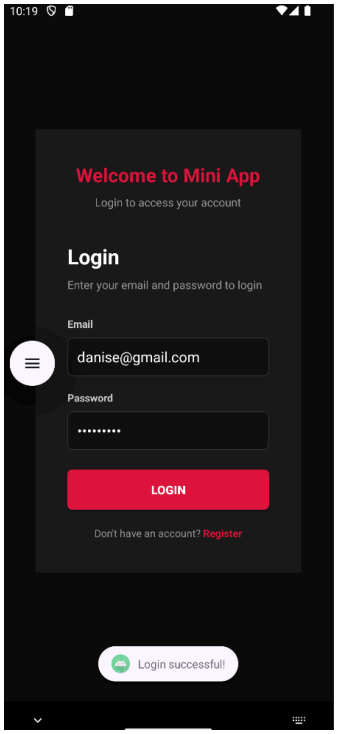
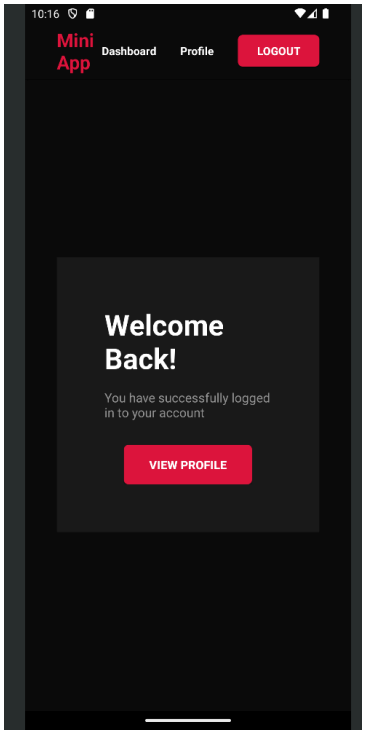
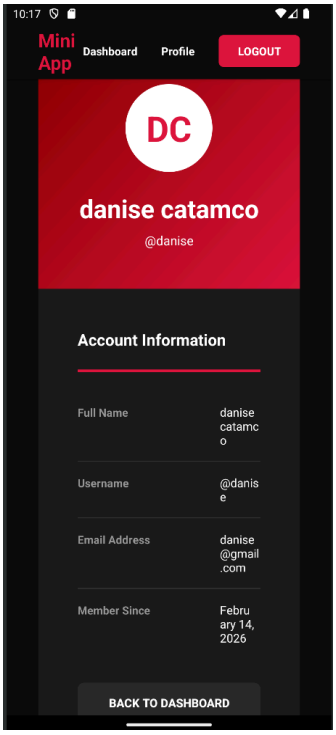
student.cituwits.com
All Bookmarks
Mini App
Dashboard
Profile
Logout
DC
danise catamco
@danise
Account Information
Full Name
danise catamco
Username
@danise
Email Address
danise@gmail.com
Member Since
February 15, 2026
Back to Dashboard

Mobile Screenshots:









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